Cell Structures

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Key cell structures include cell membrane, cell wall, nucleus, ribosomes, cytoplasm, mitochondria, chloroplasts, and vacuoles, each with a distinct function.

Round 1 - Retrieve It

1. What does the cell membrane do?

2. What does the nucleus do?

3. What do ribosomes do?

4. Which two structures are strongly associated with plant cells and not animal cells?

Round 2 - Sketch It

Draw a simple plant cell and animal cell. Label membrane, nucleus, cytoplasm, mitochondria, ribosomes, and vacuole. Add wall and chloroplasts to the plant cell.

Cell Structures - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A cell needs to make many proteins. Which structure is directly responsible?

2. A leaf cell captures light energy. Which structure is central to that job?

3. A muscle cell needs large amounts of usable energy. Which organelle would likely be abundant?

Round 4 - Reason from Evidence

Compare plant and animal cells using at least two similarities and two differences.

Cell Structures - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A **Articulate**
Explain Cell Structures in your own words. Use one example that is not printed on the cue card.

C **Connect**
Connect cell structure to function. Why would different cell types have different numbers of mitochondria or chloroplasts?

E **Extend**
A mystery cell has a cell wall, chloroplasts, and a large vacuole. Identify the cell type and defend your claim.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Cell Structures

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Controls what enters and leaves the cell; forms a boundary.

2. Contains genetic material and helps direct cell activities.

3. Build proteins.

4. Cell wall and chloroplasts; plants also often have a large central vacuole.

Sketch It - look for

Correct placement does not need microscopic precision. Plant should include cell wall and chloroplasts; both cells include membrane, nucleus, cytoplasm, mitochondria, ribosomes, vacuoles.

Apply It

1. Ribosomes.

2. Chloroplasts.

3. Mitochondria.

Reason from Evidence - look for

Similarities can include membrane, nucleus, cytoplasm, mitochondria, ribosomes, vacuoles. Differences: plant cell wall, chloroplasts, typically larger central vacuole and more rigid shape.

ACE - Extend look-for

Plant cell; those structures are characteristic of plant cells.

Suggested use	Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.